

You are required to implement **at least TWO advanced database features** in your project according to the selected database technology.

General Requirements:

For every advanced feature selected, students must provide:

1. Feature Explanation

- What the feature is
- Why it is used
- Benefits of using it
- Real-world use cases

2. Implementation

- Configure or implement the feature in their project
- Show queries/code/configuration
- Demonstrate working screenshots/output

3. Documentation

Include:

- Steps performed
- Code snippets
- Screenshots
- Challenges faced (if any)

Database-Wise Advanced Features:

1. PostgreSQL Advanced Features

- Indexing
- Transactions
- Triggers
- Partitioning
- Full-Text Search

2. SQLite Advanced Features:

- Transactions
- Triggers
- Indexing
- Full-Text Search
- Cloud/Distributed Deployment

3. MongoDB Advanced Features

- Indexing
- Replication
- Sharding
- GridFS
- Transactions
- Full-Text Search

4. Firebase Advanced Features

- Real-Time Synchronization
- Cloud Deployment
- Authentication + Security Rules
- Transactions
- Offline Persistence

5. Apache Cassandra Advanced Features:

- Partitioning
- Replication
- Distributed Deployment
- Indexing
- Time-Series Modeling

6. Neo4j Advanced Features:

- Graph Traversal
- Recommendation Systems
- Indexing
- Pathfinding Algorithms
- Graph Analytics

Important

- Simply mentioning a feature is NOT sufficient.
- Features must be practically implemented.

- Screenshots and code evidence are mandatory.
- Students are encouraged to select features relevant to their project domain.

Submission:

1. Submit a single ZIP file including:
 ProjectName_StudentID/
 — Report.pdf
 — Code/
 — Screenshots
2. Report (Required): For EACH of the 2 advanced features:
 - Feature name
 - What it is + why used (short)
 - Benefits
 - Query/code implementation
 - Output explanation
 - Screenshots (mandatory)
3. Code
 - All database queries
 - Configuration files (if any)
4. Screenshots (Mandatory)
 - Include:
 - Query execution
 - Output results
 - Proof feature is working